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The effect of common ownership among supply chain parties on decision-making and surplus with manufacturer encroachment

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ABSTRACT

In the financial economics field, several studies assess the effect of common ownership on the economic consequence, decision-making of firms, and social performance. In this situation, based on recent supply chain practices and manufacturer encroachment, we examine the effect of common ownership on supply chain parties composed of a retailer and a manufacturer and social performance. From our model analysis, we demonstrate that manufacturer encroachment improves the retailer's profit under specific economic environments, while generally, manufacturer encroachment harms the retailer's profit because the manufacturer obtains market share through a direct channel. Additionally, we demonstrate that increasing common ownership improves total surplus in a specific case. Therefore, our analysis does not only propose the positive effects of common ownership among all vertical supply chain parties for supply chain coordination, as opposed to what is indicated by the real world, but also implies that society should accommodate common ownership among supply chain parties in specific economic environments from the perspective of total surplus.

1. Introduction

In the financial economics field, many practitioners, investors, and researchers shed light on common ownership among competing firms. For example, in Financial Times, it is stated as "Global regulators are starting to home in on an academic theory on the drawbacks of overlapping corporate ownership by investors, a trend that could pose a threat to the $80tn asset management industry".[1] This is one of the typical explanations of common ownership and one can recognize the attention to common ownership by stakeholders and regulators. Additionally, according to Azar et al. (2018) and Dennis et al. (2022), one can confirm the importance of common ownership for the firm value and social performance by the stakeholders and researchers. In particular, the relationship between product market competition and common ownership is widely investigated in the financial economics field. This is because while common ownership is one of the related issues in the finance field, from the definition of common ownership, product market competition plays an important role to discuss the effect of it.

In this study, we also focus on common ownership among competing firms in the product market. In particular, we shed light on the asymmetric competition among supply chain parties. When shareholders become increasingly powerful, supply chain participants are often mutually owned by the same institutional owners who share common ownership. For example, Vanguard Group holds

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5.5% shares of Rollins, Inc.[2] and 8.08% shares of Dick's Sporting Goods.[3] Additionally, according to Chen (2021), Tenneco Inc., a designer and manufacturer of emission control products, shares common ownership with customers (e.g., Caterpillar Inc. and Harley Davidson Inc.). These examples illustrate interesting aspects of common ownership along supply chains regarding a company's portfolio holdings. However, while common ownership is considered a social issue from the perspective of antitrust policies, previous studies have failed to examine the impact of common ownership on supply chain participants, assuming a dual-channel supply chain. Therefore, this study analytically explores the effect of common ownership on decision-making, the economic consequence of supply chain parties, and social performance.

IT developments have made it easier for manufacturers to open a direct channel, leading to manufacturer encroachment. For example, Hewlett-Packard Co. has a direct channel for selling its products to consumers without using retail stores. Additionally, reputed companies such as Fast Retailing Co., Ltd., Procter and Gamble, and Apple Inc. use direct channels to sell their products. With manufacturer encroachment, the manufacturer becomes a retailer's competitor rather than a supply chain partner. Specifically, it becomes challenging to consider the ownership of supply chain parties if the owner is a partner or a competitor. For example, the Vanguard Group holds an 11.50% share of Best Buy[4] and 7.4% of General Electric Co..[5] This example represents the case in which the retailer (Best Buy) is encroached by the manufacturer (General Electric) through common ownership. How do manufacturers' and retailers' decision-making change in the threat of manufacturer encroachment? This study considered the following research questions: (i) What is the effect of manufacturer encroachment and common ownership on supply chain parties' decision-making? (ii) What is the effect of manufacturer encroachment and common ownership on retailer profits? (iii) What is the surplus effect of common ownership?

This study constructs an analytical model based on game theory to explore the impact of common ownership on decision-making processes and surplus management. In our model, the supply chain comprises one retailer and one manufacturer. The manufacturer produces goods and sells them to the retailer at wholesale price. The retailer supplies the product to the final goods market by determining its quantity. The manufacturer can also decide whether to encroach on the final goods market. When a manufacturer encroaches quantity competition among the retailer and manufacturer occurs in the retail market. Our model assumes common ownership among firms to emphasize the other parties' profit. The timeline of these events is as follows: First, the manufacturer chooses whether to encroach. Second, the manufacturer decides on the wholesale price. Finally, in the non-encroachment scenario, only the retailer determines the quantity, whereas, in the encroachment scenario, the retailer and manufacturer determine the quantity. We describe our basic model in Section 3.[6]

From our analysis, we obtained the following results. In the non-encroachment case, the degree of channel substitution does not affect the retailer's profit, because the competition does not occur between the participants in the supply chain. In the non-encroachment case, the wholesale price charged by the manufacturer increases as the degree of common ownership increases. Under large common ownership, while the retailer raises the supplied quantity to enhance the manufacturer's profit, anticipating retailer behavior, the manufacturer raises the wholesale price to avoid excessive supply in the retail market. In the case of encroachment, the degree of channel substitution plays an important role in considering our result. Under a large degree of channel substitution, it is not profitable for the manufacturer and retailer to supply large quantities in the retail market, because intense competition leads to cannibalization. Therefore, the manufacturer manages the retail market to enhance its profit and the retailer's profit under common ownership. From the discussion presented above, the following two effects are important when considering the effect of common ownership on wholesale prices: (i) Increasing wholesale prices because the retailer raises quantity by emphasizing the manufacturer's profit, and (ii) decreasing wholesale prices to provide the market share to retailers. In the case of a small degree of channel substitution, the wholesale price increases and the retailer increases the quantity as the degree of common ownership increases. Therefore, each firm improves its profit under common ownership by increasing its profit. In this case, one can infer that consumer surplus may decrease with increasing common ownership, because large common ownership leads to tacit collusion in the product market by supply chain parties. However, this is not always true in our model. In the case of a small degree of channel substitution, when the wholesale price increases, although the quantity supplied by the manufacturer decreases with increasing common ownership, that supplied by the retailer increases. The retailer radically increases the quantity in the retail market by reducing the quantity supplied by the manufacturer. Consequently, the total quantity supplied to the retail market increases as common ownership increases. In our additional analysis, we check the robustness of this result by assuming a direct selling cost for the manufacturer in Section 5.4. In addition, by using a numerical example, we confirm that our main results are presented in Section 5.2.

These results represent important contributions to supply chain management practices and earlier studies from the perspective of decision-making and policy-making. First, our results on retailer profits suggest that common ownership improves a retailer's profit. This also indirectly enhances the profit in the encroachment case. These findings imply that retailers should aggressively join the

[2] We obtain this information from the Rollins 2020 Annual Report and 2021 Proxy Statement, p.4. URL: https://www.rollins.com/~/media/Files/R/Rollins-V2/documents/annual-reports/2020-rollins-annual-report-v1.pdf (Last accessed, September 12, 2021)

[3] We obtain this information from the Notice of Dick's Sporting Goods' Annual Meeting of Stockholders and 2021 Proxy Statement, p.5. URL: https://s27.q4cdn.com/812551136/files/doc_financials/2021/04/DEF-14A-2021-DKS-Proxy-Statement_FINAL.2.pdf (Last accessed, September 12, 2021)

[4] We obtain this information from the 2021 Proxy statement of Best Buy, p.40. URL: http://s2.q4cdn.com/785564492/files/doc_financials/2021/ar/2021-Best-Buy-Proxy.pdf (Last accessed, September 12, 2021)

[5] We obtain this information from the 2021 Notice of Annual Meeting and Proxy Statement of General Electric Co., p.25. URL: https://www.ge.com/sites/default/files/ge_proxy2021.pdf (last accessed September 12, 2021)

[6] Note that, in additional analysis, we assume an additional marginal cost when the manufacturer sells products to consumers directly.

common ownership between retailers and manufacturers for supply chain coordination as a managerial insight. In particular, when the degree of channel substitution is small and the degree of common ownership is large, manufacturer encroachment improves retailer profit. Therefore, the practical implication of this result for supply chain management is that shareholders should increase their degree of common ownership under a small degree of channel substitution. Second, we propose optimal decision-making by supply chain parties with common ownership. For optimal decision-making, it is important to support supply chain parties in practice. Third, we find that common ownership has positive effects on the total surplus in the encroachment case. Hence, regarding supply chain management, society should accommodate common ownership by manufacturers and retailers from the perspective of an antitrust policy.

2. Related literature

Our contributions can be clarified through the earlier supply chain and common-ownership studies. Additionally, using the game theoretical approach, many supply chain management studies elucidate the effects of opening a direct channel by a manufacturer (e.g., Chiang et al., 2003; Hamamura & Zennyo, 2021; Hsiao & Chen, 2013; Matsui & Tsunoda, 2022; Matsushima & Mizuno, 2018; Mizuno, 2012; Tsay & Agrawal, 2004). In this context, Chiang et al. (2003) presents a traditional economic analysis of a dual-channel supply chain. Additionally, Hamamura and Zennyo (2021), Pei and Yan (2015), Yoon (2016), and Zhang et al. (2020) use non-cooperative game theory to demonstrate the positive effects of spillover investments. From a managerial perspective, Matsui and Tsunoda (2022) considers the effects of exclusive territories on supply chain management. Additionally, Hua et al. (2010) includes the assumption of a delivery lead-time for a manufacturer in a direct channel. Few studies claim that manufacturer encroachment positively affects a retailer's profit. Only a few studies have examined the effects of common ownership on supply chain management. We focus on the study that is closest to our model.

The effects of the dual-channel on the retailer's profit and surplus were also investigated in previous studies (Arya et al., 2007; Sun et al., 2019; Yoon, 2016; Zhang et al., 2020). Arya et al. (2007) demonstrates the bright side of manufacturer encroachment, which enhances retailers' profit. Additionally, Yoon (2016) describes that a manufacturer's cost-reducing investment improves the retailer's profits through the spillover effects of its investment. Furthermore, Zhang et al. (2020) considers demand-enhancing investments. Pei and Yan (2015) conducts supportive retailers' services that coordinate the dual-channel distribution as an effective incentive. They assume that additional retail services positively impact demand. Hamamura and Zennyo (2021) describes the effects of a retailer's voluntary investment in cost reduction on manufacturer encroachment. In addition, Hsiao and Chen (2013) explores an asymmetric channel structure despite the ex-ante symmetry between manufacturers.

In the supply chain study, Aviv and Shamir (2021) is closest to our research. They also explore the impact of common ownership on supply chain parties using non-cooperative game theory. However, while we assume common ownership between a manufacturer and retailer, Aviv and Shamir (2021) considers common ownership between horizontal retailers. In other words, while we shed light on the impact of common ownership on the strategies of supply chain parties with supplier encroachment, Aviv and Shamir (2021) focus only on indirect channels through retailers, assuming that one manufacturer does not open the direct channel. Additionally, while Aviv and Shamir (2021) assumes asymmetric information between supply chain parties, we do not assume this to clarify the impact of common ownership on strategies and surplus. Moreover, assuming a two-echelon supply chain comprised of one assembler and multiple upstream suppliers, Xiao et al. (2021) considers backward shareholding among supply chain parties. However, they do not examine the direct channels of the manufacturer.

Additionally, in the economics field, based on non-cooperative game theory, several findings in the study have described economic analyses of common (overlapping) ownership (e.g., Li & Zhang, 2021; López & Vives, 2019; Sato & Matsumura, 2020) because common ownership is an important issue from the perspective of antitrust policy. Li and Zhang (2021) demonstrates the adverse effects of common ownership on the consumer and total surplus in a duopoly, whereas Sato and Matsumura (2020) shows that increasing common ownership engenders decreased entry and may or may not improve total surplus. Therefore, common ownership helps improve the total surplus under a particular economic environment. Additionally, López and Vives (2019) shows that considering R&D spillover, common ownership enhances consumer surplus and total surplus. Among these reports, few have described studies demonstrating the positive aspects of common ownership on consumer surplus and total surplus with product market competition. Our study also assesses the positive effects of common ownership.

Empirical studies have indicated the importance of common ownership in supply chain management (Chen, 2021; Freeman, 2021).[7] In the recent study, Chen (2021) claims that common ownership facilitates innovation by improving relationships between supply chain parties. Additionally, Freeman (2021) demonstrates that common ownership enhances innovative collaboration by increasing R&D, patenting, and patent cross-citations. These empirical results indicate that common ownership has positive management. While common ownership among supply chain parties influences decision-making in supply chain studies, Aviv and Shamir (2021) only examines its effects. Therefore, one can infer that some light must be shed on common ownership among supply chain parties.[8]

[7] Recently, financial economic studies empirically examine the effect of common ownership (e.g., Zhu et al., 2024). In these studies, several empirical studies analyze the effect of common ownership on the competition and decision-making of firms (e.g., Azar et al., 2018; Cheng et al., 2022; Dennis et al., 2022; Ramalingegowda et al., 2021). Therefore, one may infer that, recently, financial economics researchers widely pay attention to common ownership.

[8] One may consider that our model is close to vertical shareholdings research (e.g., Flath, 1989; Fu et al., 2018). However, these studies do not assume manufacturer encroachment.

We summarize the key differences between our study and previous studies as follows. (i) While prior studies do not consider common ownership among vertical supply chain parties with manufacturer encroachment, we consider common ownership by a shareholder between a manufacturer and retailer. (ii) We demonstrate that manufacturer encroachment positively impacts retailer profit. This result has been corroborated by several studies (e.g., Arya et al., 2007; Yoon, 2016). (iii) Assuming a dual-channel supply chain, we demonstrate the positive effects of common ownership on total surplus. While several prior studies demonstrate the positive impact of common ownership on total surplus (López & Vives, 2019; Sato & Matsumura, 2020), they do not assume a supply chain. Our study finds that common ownership supports supply chain participants and contributes to the total surplus. While common ownership is profitable, it harms the total anti-competition surplus. Therefore, the impact of common ownership on supply chain parties influences decisions regarding supply chain management and anti-competitive policies. Additionally, our results provide a new explanation for the impact of common ownership on supply chain management. Thus, our results explain joint investments by shareholders among supply chains, indicating that a firm's profits improve with common ownership. Production is commonly owned in many cases (e.g., Rollins Inc. and Dick's Sporting Goods, Caterpillar Inc., Harley Davidson Inc., and Best Buy and General Electric). Our findings can explain the common ownership practices of manufacturers. Our results suggest that manufacturers must determine their levels of common ownership and strategies after observing the degree of channel differentiation. Therefore, this study not only contributes to supply chain practice from the economic analysis perspective, but also suggests important implications from the perspective of antitrust policy.

3. Main model

This study assumes a supply chain comprising a manufacturer (M) and a retailer (R) because we shed light on the dual-channel supply chain. The manufacturer produces goods at a marginal cost, c. For simplicity, this study normalizes $c = 0$. The manufacturer sells products to the retailer at a wholesale price, w, before the retailer sells them to consumers as an indirect distribution channel. In addition, we assume that the manufacturer determines whether to open a direct distribution channel. In this study, the manufacturer opens a direct channel as an encroachment case, and one where it does not open a direct channel as a non-encroachment case.

When a manufacturer encroaches on the retail market using a direct channel, quantity competition occurs between the two channels. In this case, the manufacturer and retailer select the quantities q_M and q_R, respectively. In this study, p_M and p_R represent the prices in the direct and indirect channels, respectively. Consumers can buy products from the direct and indirect distribution channels. Following prior studies (e.g., Dixit, 1979), this study assumes the utility function of a representative consumer to be

$$U(q_R, q_M) = a(q_R + q_M) - \frac{1}{2}\left(q_R^2 + 2\theta q_R q_M + q_M^2\right), \tag{1}$$

where $a(> 0)$ represents the potential market size and $\theta \in (0, 1]$ represents the degree of channel substitution. When θ is high, the two channels are highly substitutable. When θ approaches zero, both channels engage in a monopolistic market.

Considering the utility maximization of the consumer, subject to the budget constraint, the following inverse demand functions are obtained:

$$p_i = a - q_i - \theta q_j, \quad i, j = R, M, \ i \neq j. \tag{2}$$

Based on Singh and Vives (1984), this study assumes that consumer surplus is $CS = (q_R^2 + 2\theta q_R q_M + q_M^2)/2$.

We define the profit functions of the retailer and manufacturer as:

$$\pi_R = (p_R - w)q_R, \tag{3}$$

$$\pi_M = wq_R + p_M q_M, \tag{4}$$

In our main model, while we do not assume the additional selling cost in the direct channel (*i.e.,* , direct channel inefficiencies) for simplicity, we will discuss the impact of direct selling cost, which was proposed by Arya et al. (2007) in Section 5.4. In this study, the total surplus is defined as the sum of the profits of all firms and consumer surpluses: $TS = CS + \pi_R + \pi_M$. Such economic analysis is typically applied to policy-making. Based on an economic analysis, an antitrust policy in e-commerce was discussed. In this case, consumer and total surpluses are important variables to consider when discussing policy-making in the distribution. Therefore, our assumption of the surplus is reasonable when considering supply chain or distribution practices.

When the manufacturer does not open a direct channel, we assume $q_M = 0$. Hence, the inverse demand function of an indirect channel is $p_R = a - q_R$. In this case, the manufacturer's profit function is $\pi_M = wq_R$: Additionally, the manufacturer supplies the product through the retailer (i.e., an indirect channel). In this study, we assume that the manufacturer can choose to open the direct channel. In other words, we exclude cases in which the manufacturer does not supply products through indirect channels. This is because if the manufacturer has the option to stop the trade with the retailer, then the manufacturer sets $q_M = (a - c)/2$ and earns $\pi_M = V_M = (a - c)^2/4$ as a monopolist profit. In this case, the manufacturer always closes the indirect channel, and therefore, we assume that the manufacturer must open the indirect channel in our analysis because we will discuss the common ownership among supply chain parties with the existence of both firms.

Based on the results of previous studies, this study assumes common ownership between a manufacturer and a retailer (e.g., Li & Zhang, 2021; López & Vives, 2019; Sato & Matsumura, 2020). In this case, the profit functions of firms V_M and V_R are:

$$V_M = \pi_M + \lambda\pi_R, \tag{5}$$

$$V_R = \pi_R + \lambda\pi_M, \tag{6}$$

where $\lambda \in (0, 1)$ denotes the degree of ownership. Because earlier studies assumed symmetric shares of ownership by owners, we also assumed symmetric shares, λ, for simplicity. In our main model, we assume a symmetric ownership share to simplify our discussion and relax this assumption in Section 5.3 to check the robustness of our result. However, if we obtain an important result under the specific assumption, we can also show that in the relaxed case. When λ is large, the common owners share a large share of each firm. Our assumption is unique in terms of dual-channel supply chain studies because we aim to investigate the impact of common ownership on strategies, profits, and surpluses. In previous sections, using practical examples and previous studies, we discuss the importance of exploring the impact of common ownership on supply chains. Therefore, λ is the most important assumption in examining the research question. In the main analysis, we assume $\lambda < \sqrt{2} - 1$ to ensure the positive outcomes.

We assume the following timing for the game. First, the manufacturer determines whether to open a direct channel (stage 1). Second, the manufacturer sets the wholesale price (stage 2). Third, quantity competition occurs in the product market (Stage 3). In particular, when the manufacturer opens a direct channel, the manufacturer and retailer decide on quantities; otherwise, only the retailer decides on the supplied quantity. We assume that the manufacturer decides on the channel strategy because, in practice, whether the manufacturer opens a direct channel is a long-term decision. We use the sub-game perfect equilibrium as an equilibrium concept in this study and analyze the equilibrium by backward induction.

4. Equilibrium analysis

We consider the following two cases (i.e., non-encroachment and encroachment), given the encroachment decision in stage 1. In Section 4.1, we propose the outcome of decisions after the manufacturer decides not to open a direct channel. In Section 4.2, we propose the outcomes of the encroachment case. Finally, we analyze the decision of the manufacturer on whether to open a direct channel in Section 4.3.

4.1. Non-encroachment case

As described in this subsection, we derive the outcomes for the non-encroachment case. We denote the outcomes in this case with superscript NE. In this case, the retailer chooses quantity at stage 3 to maximize Eq. (6) under $q_M = 0$. We identify the retailer's quantity decision in stage 3 as follows under the non-encroachment case.

$$q_R^{NE} = \frac{a - w(1-\lambda)}{2}. \tag{7}$$

This outcome reflects the effects of common ownership. When common ownership is large, q_R^{NE} is large because it decreases because, in this case, the retailer emphasizes the manufacturer's profit and induces profit from wholesales for the manufacturer. Therefore, common ownership modifies the double marginalization problem through its effects.

Next, we consider the manufacturer's wholesale price decisions. Using Eq. (7), the manufacturer selects w to maximize Eq. (5). Therefore, we obtain the following outcome.

$$w^{NE} = \frac{a}{(1-\lambda)(2+\lambda)}. \tag{8}$$

Based on this outcome, we consider the effects of common ownership on wholesale prices in the supply chain. Differentiating w^{NE} by λ, we obtain

$$\frac{\partial w^{NE}}{\partial \lambda} = \frac{a(1+2\lambda)}{(1-\lambda)^2(2+\lambda)^2} > 0. \tag{9}$$

As shown in Eq. (9), the wholesale price increases as common ownership increases. This is true because in this case, the manufacturer wishes to raise the retailer's profit, preventing excessive supply by the retailer, although the retailer wishes to raise the manufacturer's profit by increasing the quantity. Excessive supply is the most important problem in cases of double marginalization. From Eq. (7), increasing the wholesale price is the best method to alleviate this problem. Therefore, the manufacturer increases the wholesale price to mitigate excessive supply in the quantity decision.[9] From these outcomes, we obtain the following proposition.

Proposition 1. *In the non-encroachment case, the wholesale prices increase as the degree of common ownership increases.*

Using Eqs. (7) and (8), we obtain the following outcomes in this case,

$$q_R^{NE} = \frac{a(1+\lambda)}{2(2+\lambda)}, \qquad p_R^{NE} = \frac{a(3+\lambda)}{2(2+\lambda)},$$
$$\pi_R^{NE} = \frac{a^2(1+\lambda)\left(1-2\lambda-\lambda^2\right)}{4(1-\lambda)(2+\lambda)^2}, \qquad \pi_M^{NE} = \frac{a^2(1+\lambda)}{2(1-\lambda)(2+\lambda)^2},$$
$$V_R^{NE} = \frac{a^2(1+\lambda)^2}{4(2+\lambda)^2}, \qquad V_M^{NE} = \frac{a^2(1+\lambda)^2}{4(2+\lambda)}.$$

[9] It can be inferred that the manufacturer decreases the wholesale price to mitigate the double marginalization problem under the price decision.

When $\lambda < \sqrt{2} - 1$, all outcomes are positive, and the second order conditions are satisfied.

From this outcome, we consider the effects of λ on q_R^{NE} to be

$$\frac{\partial q_R^{NE}}{\partial \lambda} = \frac{a}{2(2+\lambda)^2} > 0. \tag{10}$$

The quantity set by the retailer in the non-encroachment case increases with λ. This outcome implies that when firms emphasize the other supply chain party's profit, the retailer increases the quantity because the manufacturer raises the wholesale price. In this case, the retailer also emphasizes the manufacturer's profit and increases its quantity to improve the manufacturer's profit under a high wholesale price. From this outcome, we obtain the following proposition.

Proposition 2. *In the non-encroachment case, the quantity set by the retailer increases with the degree of common ownership.*

This proposition is influenced by the effects of common ownership on wholesale prices. Therefore, firms' common ownership has an important effect on the decision-making of the retailer and manufacturer.

Finally, we obtain the following consumer surplus and total surplus in the non-encroachment case.

$$CS^{NE} = \frac{a^2(1+\lambda)^2}{8(2+\lambda)^2}, \tag{11}$$

$$TS^{NE} = \frac{a^2\left(7+10\lambda+3\lambda^2\right)}{8(2+\lambda)^2}. \tag{12}$$

In addition, we analyze the effects of λ on CS^{NE} and TS^{NE}.

$$\frac{\partial CS^{NE}}{\partial \lambda} = \frac{a^2(1+\lambda)}{4(2+\lambda)^3} > 0, \tag{13}$$

$$\frac{\partial TS^{NE}}{\partial \lambda} = \frac{a^2(3+\lambda)}{4(2+\lambda)^3} > 0. \tag{14}$$

This result indicates that CS^{NE} and TS^{NE} increase as λ increases. This result is interesting because few studies demonstrate the positive effects of common ownership on consumer surplus and total surplus with product market competition (López & Vives, 2019; Sato & Matsumura, 2020). When the manufacturer emphasizes the retailer's profit, it improves. Additionally, in the non-encroachment case, increasing λ enhances the quantity supplied in the product market. Therefore, these positive effects engender an improvement in the total surplus in the non-encroachment scenario. From this analysis, we obtain the following proposition.

Proposition 3. *In the non-encroachment case, increasing λ improves the total surplus; that is, $\partial TS^{NE}/\partial\lambda > 0$.*

In earlier studies, an increase in the share of ownership was demonstrated to harm the total surplus, because common ownership engenders cartelization (e.g., Li & Zhang, 2021). In addition, earlier studies examined the anti-competitive effects of common ownership and antitrust regulations (e.g., Azar et al., 2018). Therefore, this result demonstrates new findings related to common ownership.

4.2. Encroachment case

As explained in this subsection, we derive the outcomes of the encroachment case. We present the outcomes, in this case, using the superscript E. In this case, the retailer and manufacturer decide on the quantity at stage 3 to maximize Eqs. (5) and (6). The quantity decisions made by the retailer and manufacturer in Stage 3 are identified as

$$q_R^E = \frac{a\left[2-\theta(1+\lambda)\right] - 2(1-\lambda)w}{4-\theta^2(1+\lambda)^2}, \tag{15}$$

$$q_M^E = \frac{a\left[2-\theta(1+\lambda)\right] + \theta(1-\lambda)(1+\lambda)w}{4-\theta^2(1+\lambda)^2}. \tag{16}$$

From this outcome, when the wholesale price increases, q_R^E decreases and q_M^E increases.[10]

Next, we consider Stage 2. At this stage, the manufacturer chooses the wholesale price to maximize $V_M(q_R^E, q_M^E)$. We get

$$w^E = \frac{a\left[\theta^3(1+\lambda)^3 - 2\theta^2(2-\lambda)(\lambda+1)^2 - 4\theta\lambda(1+\lambda) + 8\right]}{2(1-\lambda)\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]}. \tag{17}$$

Based on this outcome, we consider the impact of common ownership on wholesale prices in the supply chain in an encroachment case. Differentiating w^E by λ, we obtain

$$\frac{\partial w^E}{\partial \lambda} = \frac{a(1-\theta)\left[3\theta^4(1+\lambda)^4 - 4\theta^2\left(5+\lambda^2\right)(1+\lambda)^2 + 32\lambda + 16\right]}{(1-\lambda)^2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2}, \tag{18}$$

[10] From the first-order conditions of each firm's strategies, λ only has a role in shifting the slope of the best-response functions. Therefore, a strategic substitute between quantities holds in this case.

where $3\theta^4(1+\lambda)^4 - 4\theta^2(5+\lambda^2)(1+\lambda)^2 + 32\lambda + 16 > 0$ holds and $\partial w^E/\partial\lambda > 0$. This result implies that the wholesale price increases as λ increases under small θ. This implication corresponds to the non-encroachment case because when θ is sufficiently small, each channel approaches a monopoly, and the situation is similar to the non-encroachment case. Therefore, the impact of λ on w^E is the same as that in the non-encroachment case. However, under a large θ, the effect of increasing λ differs from that in the non-encroachment case (i.e., $\partial w^E/\partial\lambda < 0$). Under a large θ, it is not profitable for the manufacturer to supply large quantities in the retail market because intense competition leads to cannibalization. Therefore, the manufacturer handles the retail market to enhance its own profit and the retailer's profit under common ownership. From the discussion presented above, the following two effects are important to consider the effect of λ on wholesale price: (i) increasing wholesale prices because the retailer raises quantity by emphasizing the manufacturer's profit, and (ii) decreasing wholesale prices to give the market to retailers. The second effect plays an important role when θ is large. Based on this outcome, we obtain the following proposition.

Proposition 4. *In the encroachment case, when* $3\theta^4(1+\lambda)^4 - 4\theta^2(5+\lambda^2)(1+\lambda)^2 + 32\lambda + 16 < 0$ *and* $\lambda < \sqrt{2}-1$ *hold true, the wholesale price decreases as the degree of common ownership increases.*

Using, Eq. (17), we calculate and obtain the following outcomes under an encroachment case.

$$q_R^E = \frac{2a(1-\theta)(1+\lambda)}{4(2+\lambda) - 3\theta^2(1+\lambda)^2}, \tag{19}$$

$$q_M^E = \frac{a\left[2-\theta(1+\lambda)\right]\left[4+\theta(1+\lambda)+2\lambda\right]}{2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]}, \tag{20}$$

$$p_R^E = \frac{a\left[4(3+\lambda) - 4\theta - 4\theta^2(1+\lambda)^2 + \theta^3(1+\lambda)^2\right]}{2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]}, \tag{21}$$

$$p_M^E = \frac{a\left[4(2+\lambda) - 2\theta(1-\lambda)(1+\lambda) + \theta^2\left(1+6\lambda+5\lambda^2\right)\right]}{2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]}, \tag{22}$$

$$\pi_R^E = \frac{2a^2(1-\theta)^2(1+\lambda)\left[2-(2-\theta)(2+\theta)\lambda - 2(1-\theta)(1+\theta)\lambda^2 + \theta^2\lambda^3\right]}{(1-\lambda)\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]}, \tag{23}$$

$$\pi_M^E = w^E q_R^E + p_M^E q_M^E, \tag{24}$$

$$V_R^E = \pi_R^E + \lambda\pi_M^E, \tag{25}$$

$$V_M^E = \pi_M^E + \lambda\pi_R^E. \tag{26}$$

When $\lambda < \sqrt{2}-1$ holds, all outcomes are positive, and the second order conditions are satisfied.

From this outcome, we consider the effects of λ on q_R^E.

$$\frac{\partial q_R^E}{\partial\lambda} = \frac{2a(1-\theta)\left(4+3\theta^2(1+\lambda)^2\right)}{\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2} > 0. \tag{27}$$

Therefore, the effect of λ is the same as in the non-encroachment case. When the second effect in wholesale pricing is strong, the manufacturer intends to enhance the profit of the retailer from the product market to avoid competition. In this case, the retailer emphasizes the manufacturer's profits. The increasing quantity raises the transaction quantity between the retailer and manufacturer to improve the manufacturer's profit. Therefore, $\partial q_R^E/\partial\lambda > 0$ always holds.

Additionally, considering the effects of λ on q_M^E, the mechanism is clear.

$$\frac{\partial q_M^E}{\partial\lambda} = -\frac{4a(1-\theta)\theta\left(3+4\lambda+\lambda^2\right)}{\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2} < 0. \tag{28}$$

This outcome also indicates that increasing λ lowers the quantity of manufacturers in this case. In this case, whereas the retailer raises the quantity to improve the manufacturer's profit, the manufacturer can reduce the quantity through the direct channel because the manufacturer can earn profit from the retailer's behavior. If the manufacturer supplies a large quantity, then one can infer that this behavior has led to an excessive supply in the product market. These manufacturer and retailer behaviors point to collusion by common ownership, as demonstrated in earlier economic studies.

In addition, based on the outcomes of Eqs. (27)–(28), we propose the following.

Proposition 5. *In the encroachment case, the quantity set by the retailer increases as the degree of common ownership increases.*

Proposition 6. *In the encroachment case, the quantity set by the manufacturer decreases as the degree of common ownership increases.*

In an intense competition case (i.e., a high degree of θ), because the retailer desires to improve the manufacturer's profit under a large λ, the retailer aims to increase trade to enhance the manufacturer's profit while avoiding product market competition. Therefore, under large λ, the retailer chooses a *tacit collusion* strategy. This result is the same as that obtained in previous studies.

Finally, the consumer surplus and total surplus in the encroachment case are computed as

$$CS^E = \frac{a^2 A_1}{8\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2}, \tag{29}$$

$$TS^E = \frac{a^2 A_2}{8\left[3\theta^2(\lambda+1)^2 - 4(\lambda+2)\right]^2}, \tag{30}$$

where $A_1 \equiv 16(5+6\lambda+2\lambda^2) - 16\theta\lambda(1+\lambda)(3+\lambda) - 4\theta^2(1+\lambda)(19+15\lambda+3\lambda^2-\lambda^3) + 4\theta^3(1+\lambda)^3(3+\lambda) + \theta^4(1+\lambda)^3(9+\lambda)$, and $A_2 \equiv 16(19+22\lambda+6\lambda^2) - 16\theta(1+\lambda)(3+\lambda)(4+\lambda) - 4\theta^2(1+\lambda)(25+57\lambda+25\lambda^2+\lambda^3) + 20\theta^3(1+\lambda)^3(3+\lambda) + \theta^4(1+\lambda)^3(3+11\lambda)$.

In addition, we analyze the effect of λ on CS^E and TS^E as

$$\frac{\partial CS^E}{\partial \lambda} = \frac{a^2(1-\theta)A_3}{2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^3}, \tag{31}$$

$$\frac{\partial TS^E}{\partial \lambda} = \frac{a^2(1-\theta)A_4}{2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^3}, \tag{32}$$

where $A_3 \equiv 16(1+\lambda) - 8\theta(4+11\lambda+6\lambda^2+\lambda^3) - 4\theta^2(1+\lambda)^2(2-5\lambda-\lambda^2) + 2\theta^3(1+\lambda)^2(19+8\lambda+\lambda^2) - 6\theta^4(1+\lambda)^4 - 3\theta^5(1+\lambda)^4$, and $A_4 \equiv 16(3+\lambda) - 8\theta(8+11\lambda+6\lambda^2+\lambda^3) + 4\theta^2(\lambda+1)^2(8+\lambda-\lambda^2) - 2\theta^3(1+\lambda)^2(1-16\lambda-5\lambda^2) - 12\theta^4(1+\lambda)^4 + 3\theta^5(1+\lambda)^4$. Eqs. (31) and (32) are positive when θ is sufficiently small. Thus, CS^{NE} and TS^{NE} increase as λ increases. We summarize this result in the following proposition:

Proposition 7. *In the encroachment case, the consumer surplus and total surplus increase as the degree of common ownership increases under specific economic conditions.*

Proof. See Appendix.

This is a surprising finding. Earlier studies conducted from an antitrust perspective find that when firms engage in competition in a product market, common ownership adversely affects consumer surplus and total surplus. In particular, in a marketing study, it is interesting that the anti-competitive effects of common ownership improve consumer surplus.

The effects of wholesale prices and quantities led to this result. For a small θ, while the wholesale price increases and the quantity supplied by the manufacturer decreases as common ownership increases, the quantity supplied by the retailer increases as common ownership increases. In this case, a high degree of common ownership leads to a large quantity chosen by the retailer to improve the manufacturer's profit. In addition, we can show $|\partial q_R^E/\partial\lambda| > |\partial q_M^E/\partial\lambda|$ straightforwardly. This outcome implies that the effect of increasing λ on the increase in retailer quantity is greater than the effect of increasing λ on the decrease in manufacturer quantity.

4.3. Encroachment decision and comparison of retailer profits

Using the earlier outcomes, we consider the manufacturer's channel decisions. First, we analyze $\Delta V_M \equiv V_M^E - V_M^{NE}$.

$$\Delta V_M = \frac{a^2\left[\theta^2\left(5+7\lambda+3\lambda^2\right)(1+\lambda)^2 - 8\theta(1+\lambda)(2+\lambda)^2 + 4(2+\lambda)^2\right]}{4(2+\lambda)\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]} > 0. \tag{33}$$

By this outcome, because $\Delta V_M > 0$ holds, the manufacturer always opens a direct channel.

The effects of manufacturer encroachment on retailer profits generally differ from those of the manufacturer's profit. After defining $\Delta V_R = V_R^E - V_R^{NE}$, we obtain the following outcome.

$$\Delta V_R = \frac{a^2 A_5}{4(\lambda+2)^2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2}, \tag{34}$$

where $A_5 \equiv 16\lambda(2+\lambda)^4 - 32\theta(1+\lambda)^2(2+\lambda)^2 + 4\theta^2(14-10\lambda-23\lambda^2-10\lambda^3-\lambda^4)(1+\lambda)^2(2+\lambda) + 8\theta^3\lambda(1+\lambda)^2(2+\lambda)^4 - \theta^4(9-2\lambda-11\lambda^2-5\lambda^3)(1+\lambda)^4$. In light of this finding, we consider the retailer's profit. The following proposition is obtained regarding the effect of manufacturer encroachment on the retailer's profit.

Proposition 8. *When θ is sufficiently small, the manufacturer encroachment enhances the retailer's profit. Formally, $\Delta V_R > 0$ for a sufficiently small θ.*

Proof. See Appendix.

This proposition indicates that manufacturer encroachment enhances the retailer's profit. Although some studies have demonstrated the positive effects of manufacturer encroachment (e.g., Arya et al., 2007; Yoon, 2016), our results are supported by other mechanisms described in earlier reports in the literature. This proposition shows that, when the degree of product substitution is sufficiently small, manufacturer encroachment improves the retailer's profit.

Fig. 1 illustrates the effect of θ on V_R^E and V_R^{NE} under $a = 1$. Fig. 1 shows that θ has no effect on V_R^{NE} because no competition exists between the supply chain parties. However, V_R^E decreases as θ increases, because the product market competition between

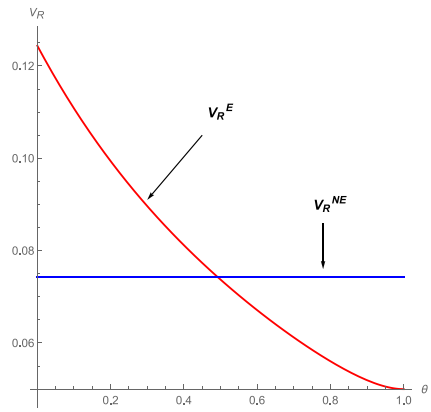

Fig. 1. Effects of θ on V_R^E and V_R^{NE} under $a = 1$ and $\lambda = 0.2$.

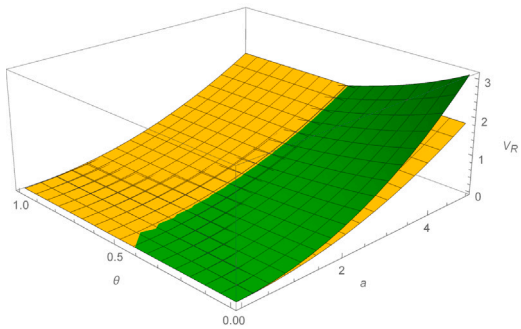

Fig. 2. Effects of θ and a on V_R^E and V_R^{NE} under $a = 1$ and $\lambda = 0.2$. Note: The yellow graph represents V_R^{NE}, and the green graph represents V_R^E.

the retailer and manufacturer intensifies as θ increases. This figure indicates that under soft competition, $V_R^{NE} < V_R^E$ holds. Under a sufficiently small θ, each firm faces the retail market as if it is a monopoly, and they earn large profits from the product market. Therefore, in the common ownership case, each firm improves its profit by improving both firms' profits.[11]

Additionally, to consider the effect of a, we visualize the 3D figure in Fig. 2. In this figure, one can observe that a does not have a serious impact on the difference of V_R in this model. When a approaches 0, the effect of θ is different from other cases. However, this case is a limited case.

Fig. 3 represents the regions where $\Delta V_R > 0$ and $\Delta V_R < 0$ under $a = 1$. This figure also suggests the existence of a specific economic condition, $\Delta V_R > 0$. In particular, under a small θ, the region of $\Delta V_R > 0$ is large. From Fig. 3, it is clear that $\lambda > 0$ plays an important role in the derivation of the results of our study. When λ is sufficiently large, $\Delta V_R > 0$ is satisfied irrespective of the degree of θ. Therefore, under a small λ, we can obtain $\Delta V_R > 0$ in a small θ, and under a sufficiently large λ, we can obtain $\Delta V_R > 0$ in any θ. From this explanation, Proposition 8 indicates the condition that $\Delta V_R > 0$ in our model.

We can confirm that while the wholesale price always increases in the non-encroachment case as common ownership increases, it decreases in the encroachment case as common ownership increases in a small degree of channel substitution. Therefore, in the case of encroachment, the retailer can enjoy profitable trade and supply large quantities in the retail market. Consequently, these positive effects lead to Proposition 8.

Because each firm, in this case, aims at *collusion*, which is likely to engender a cartel by reducing the supplied quantity, our result is similar to those reported in earlier studies based on an assumption of oligopolistic competition in a product market. However, retailers are generally exploited for their profits through manufacturer encroachment. Implying that common ownership enhances a retailer's profit through mutual help by reducing the quantity of the manufacturer in our model, our study proposes new perspectives for supply chain management.

[11] In fact, although we can demonstrate the case in which $\pi_R^E > \pi_R^{NE}$, interpreting its outcome and considering the underlying economic intuition is difficult.

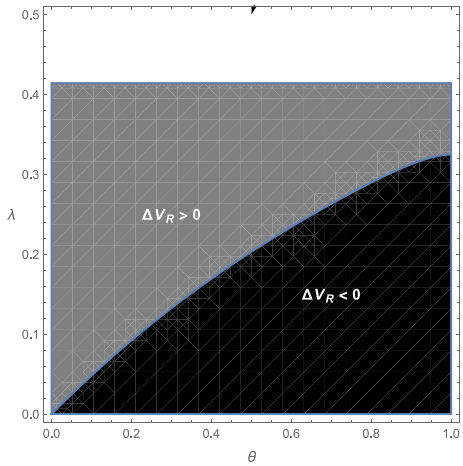

Fig. 3. Regions of $\Delta V_R > 0$ and $\Delta V_R < 0$ under $a = 1$.

Table 1
Retailer's profit matrix under specific parameters and $a = 10$.

	$\lambda = 0.05$	$\lambda = 0.1$	$\lambda = 0.15$
$\theta = 0.1$	$\Delta V_R \simeq 0.046 > 0$	$\Delta V_R \simeq 1.243 > 0$	$\Delta V_R \simeq 2.442 > 0$
$\theta = 0.2$	$\Delta V_R \simeq -0.975 < 0$	$\Delta V_R \simeq 0.181 > 0$	$\Delta V_R \simeq 1.339 > 0$
$\theta = 0.3$	$\Delta V_R \simeq -1.853 < 0$	$\Delta V_R \simeq -0.731 < 0$	$\Delta V_R \simeq 0.395 > 0$
$\theta = 0.4$	$\Delta V_R \simeq -2.616 < 0$	$\Delta V_R \simeq -1.523 < 0$	$\Delta V_R \simeq -0.425 < 0$

5. Discussion

5.1. Comparison of surplus

This section sheds light on the welfare effects. Using CS^E and CS^{NE} to elucidate the effects of encroachment on the consumer surplus, we define $\Delta CS \equiv CS^E - CS^{NE}$.

$$\Delta CS = \frac{a^2 A_6}{8(2+\lambda)^2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2} > 0, \tag{35}$$

where $A_6 \equiv 16(2+\lambda)^4 - 16\theta\lambda(1+\lambda)(2+\lambda)^2(3+\lambda) - 4\theta^2(1+\lambda)(2+\lambda)(32+31\lambda+3\lambda^2-5\lambda^3-\lambda^4) + 4\theta^3(1+\lambda)^3(2+\lambda)^2(3+\lambda) + \theta^4(1+\lambda)^3(27+13\lambda-14\lambda^2-8\lambda^3)$. This outcome implies that manufacturer encroachment always improves the consumer surplus. This is an intuitive result because if the channel increases, then the supplied quantities increase. Consequently, consumers can purchase products from their preferred channels.

In addition, using TS^E and TS^{NE} to elucidate the effects of encroachment on the total surplus, we define $\Delta TS \equiv TS^E - TS^{NE}$ as.

$$\Delta TS = \frac{a^2 A_7}{8(2+\lambda)^2\left[4(2+\lambda) - 3\theta^2(1+\lambda)^2\right]^2} > 0, \tag{36}$$

where $A_7 \equiv 48(\lambda+2)^4 - 16\theta(2+\lambda)^2(1+\lambda)(3+\lambda)(4+\lambda) - 4\theta^2(1+\lambda)(2+\lambda)(1+4\lambda+\lambda^2)(8+5\lambda+\lambda^2) + 20\theta^3(1+\lambda)^3(2+\lambda)^2(3+\lambda) - \theta^4(1+\lambda)^3(51+97\lambda+70\lambda^2+16\lambda^3)$. As $\Delta CS > 0$ is always satisfied, $\Delta TS > 0$ also holds. The results of this analysis indicate that manufacturer encroachment is profitable for both consumer and total surpluses. Therefore, consumers and regulators should accommodate the encroachment of the manufacturer into the retail market from the perspective of total surplus in the product market.

5.2. Numerical examples

In this section, we examine numerical examples to confirm whether our main results are robust or not. We specify λ and θ to calculate ΔV_R for $a = 10$. In addition, we specify the combination between λ and θ as $\lambda = 0.05, 0.1, 0.15$, and $\theta = 0.1, 0.2, 0.3, 0.4$. Because Proposition 8 requires a sufficiently small θ, we specify θ as $\theta = 0.1, 0.2, 0.3, 0.4$. In addition, from the shareholders' perspective, $\lambda \leq 0.15$ and because a large λ implies substantial domination. In this case, $\lambda < \sqrt{2} - 1$, which is the positive condition of our main outcomes, is satisfied Based on this specification, we calculate ΔV_R for nine cases and present the results in Table 1.

From Table 1, under small $\theta\,(\theta = 0.1)$, $\Delta V_R > 0$ is satisfied for any λ. However, under a sufficiently large $\theta\,(\theta = 0.4)$, $\Delta V_R < 0$ is satisfied for any λ. In addition, we discuss the effect of λ on ΔV_R, as illustrated in Fig. 3. Table 1 indicates that increasing λ increases

ΔV_R. Our numerical examples also demonstrate that ΔV_R increases with increases in θ and λ. Therefore, from this example, we can infer that when $\Delta V_R > 0$ is obtained under a sufficiently small θ, we can also obtain ΔV_R under a relatively large θ. Thus, from Proposition 8, one can confirm that our main result is robust.

5.3. Asymmetric ownership

Next, we consider the degree of asymmetry of ownership share to examine the effect of common ownership. In this case, we define the degree of ownership share between the two parties as λ_M and λ_R. Therefore, we assume that the profit function of the manufacturer is $\hat{V}_M = \pi_M + \lambda_M \pi_R$ and the profit function of the retailer is $\hat{V}_R = \pi_R + \lambda_R \pi_M$, where $\hat{}$ denotes the case of the asymmetric degree of ownership share. In this section, we assume $a = 1$ for simplicity.

By backward induction, we derive the outcomes in the equilibrium under the non-encroachment case and obtain the following result.

Result 1. *In the non-encroachment case, we obtain the following outcomes in equilibrium:*

$$\hat{q}_R^{NE} = \frac{1-\lambda_R\lambda_M}{2(2-\lambda_R\lambda_M+\lambda_M)}, \quad \hat{w}^{NE} = \frac{1-\lambda_M}{(1-\lambda_R)(2-\lambda_M-\lambda_M\lambda_R)},$$
$$\hat{\pi}_R^{NE} = \frac{(1-\lambda_M\lambda_R)\left[1+\lambda_M\lambda_R^2-(3-\lambda_M)\lambda_R\right]}{4(1-\lambda_R)(2-\lambda_M-\lambda_M\lambda_R)^2}, \quad \hat{\pi}_M^{NE} = \frac{(1-\lambda_M)(1-\lambda_M\lambda_R)}{2(1-\lambda_R)(2-\lambda_M-\lambda_M\lambda_R)^2},$$
$$\hat{V}_R^{NE} = \frac{(1-\lambda_M\lambda_R)^2}{4(2-\lambda_M-\lambda_M\lambda_R)^2} \quad \hat{V}_M^{NE} = \frac{(1-\lambda_M\lambda_R)^2}{4(1-\lambda_R)(2-\lambda_M-\lambda_M\lambda_R)},$$
$$\hat{CS}^{NE} = \frac{(1-\lambda_M\lambda_R)^2}{8(2-\lambda_M-\lambda_M\lambda_R)^2}, \quad \hat{TS}^{NE} = \frac{7-2\lambda_M(2+5\lambda_R)+\lambda_M^2\lambda_R(4+3\lambda_R)}{8(2-\lambda_M-\lambda_M\lambda_R)^2}.$$

When $0 < \lambda_M < 1$ *and* $0 < \lambda_R < (3/\lambda_M - \sqrt{(9-\lambda_M)(1-\lambda_M)/\lambda_M^2} - 1)/2 \equiv \bar{\lambda}_R$ *are satisfied, all outcomes are positive.*

Using Result 1, we consider the impact of ownership share on the quantity, wholesale price, profits, and surplus. Differentiating the variables by λ_R and λ_M, we obtain the following proposition in the asymmetric ownership share under the non-encroachment case.

Proposition 9. *The degree of ownership share has the following impact on outcomes in the non-encroachment case:*

$$\frac{\partial \hat{q}_R^{NE}}{\partial \lambda_R} < 0, \quad \frac{\partial \hat{q}_R^{NE}}{\partial \lambda_M} > 0, \quad \frac{\partial \hat{w}^{NE}}{\partial \lambda_R} > 0, \quad \frac{\partial \hat{w}^{NE}}{\partial \lambda_M} < 0,$$
$$\frac{\partial \hat{\pi}_R^{NE}}{\partial \lambda_R} < 0, \quad \frac{\partial \hat{\pi}_R^{NE}}{\partial \lambda_M} > 0, \quad \frac{\partial \hat{\pi}_M^{NE}}{\partial \lambda_R} > 0, \quad \frac{\partial \hat{\pi}_M^{NE}}{\partial \lambda_M} < 0,$$
$$\frac{\partial \hat{CS}^{NE}}{\partial \lambda_R} < 0, \quad \frac{\partial \hat{CS}^{NE}}{\partial \lambda_M} > 0, \quad \frac{\partial \hat{TS}^{NE}}{\partial \lambda_R} < 0, \quad \frac{\partial \hat{TS}^{NE}}{\partial \lambda_M} > 0.$$

Proof. See Appendix.

Proposition 9 implies that the impact of ownership share on the outcomes differs between the two parties. Therefore, in the main model, one effect dominates the other's outcomes. Specifically, the wholesale price, quantity, consumer surplus, and total surplus are improved by increasing λ. This means that when we consider the quantity, consumer surplus, and total surplus, the impact of λ_M plays an important role; when we consider the wholesale price, λ_R has a significant impact. Of course, when the supply chain party emphasizes the rival's profit, its own profit decreases, and the other party's profit increases.

Next, we focus on encroachment. In a similar manner, we derive the outcomes in this case and obtain the following result.

Result 2. *In the encroachment case, we obtain the following outcomes in equilibrium:*

$$\hat{q}_R^E = \frac{2(1-\theta)(1-\lambda_M\lambda_R)}{4(2-\lambda_M-\lambda_M\lambda_R)-\theta^2(1+\lambda_M)(3-3\lambda_M\lambda_R-\lambda_M+\lambda_R)},$$
$$\hat{q}_M^E = \frac{4(2-\lambda_M-\lambda_M\lambda_R)-2\theta(1+\lambda_M)(1-\lambda_M\lambda_R)-\theta^2(1-\lambda_M)(1+\lambda_M)(1+\lambda_R)}{2\left[4(2-\lambda_M-\lambda_M\lambda_R)-\theta^2(1+\lambda_M)(3-3\lambda_M\lambda_R-\lambda_M+\lambda_R)\right]},$$
$$\hat{w}^E = \frac{8(1-\lambda_M)-4\theta\left(2\lambda_R-\lambda_M-\lambda_M\lambda_R^2\right)-2\theta^2 B_1+\theta^3(1-\lambda_M)(1+\lambda_M)(1+\lambda_R)^2}{2(1-\lambda_R)\left[4(2-\lambda_M-\lambda_M\lambda_R)-\theta^2(1+\lambda_M)(3-3\lambda_M\lambda_R-\lambda_M+\lambda_R)\right]},$$
$$\hat{\pi}_R^E = \frac{2(1-\theta)^2(1-\lambda_M\lambda_R)\left[2-\lambda_R\left(6-2\lambda_M-\theta^2(1-\lambda_M)(1+\lambda_M)\right)+\lambda_R^2\left(2\lambda_M+\theta^2(1-\lambda_M)(1+\lambda_M)\right)\right]}{(1-\lambda_R)\left[4(2-\lambda_M-\lambda_M\lambda_R)-\theta^2(1+\lambda_M)(3-3\lambda_M\lambda_R-\lambda_M+\lambda_R)\right]^2},$$
$$\hat{\pi}_M^E = \frac{B_2}{4(1-\lambda_R)\left[4(2-\lambda_M-\lambda_M\lambda_R)-\theta^2(1+\lambda_M)(3-3\lambda_M\lambda_R-\lambda_M+\lambda_R)\right]^2},$$

$$\hat{V}_R^E = \hat{\pi}_R^E + \lambda_R \hat{\pi}_M^E, \quad \hat{V}_M^E = \hat{\pi}_M^E + \lambda_M \hat{\pi}_R^E,$$
$$\hat{CS}^E = \frac{\left[4(1-\theta)\theta(1-\lambda_M\lambda_R) + B_3\right]^2}{8\left[4(2-\lambda_M-\lambda_M\lambda_R) - \theta^2(1+\lambda_M)(3-3\lambda_M\lambda_R-\lambda_M+\lambda_R)\right]^2},$$
$$\hat{TS}^E = \hat{CS}^E + \hat{\pi}_R^E + \hat{\pi}_M^E,$$

where $B_1 \equiv 2+\lambda_M(1-\lambda_R)^2 - \lambda_M^2(1+2\lambda_R-\lambda_R^2)$, $B_2 \equiv 16(6+\lambda_M^2(1+3\lambda_R-\lambda_R^2-\lambda_R^3) - \lambda_M(6+2\lambda_R-4\lambda_R^2)-4\lambda_R) - 64\theta(1-\lambda_M)(1-\lambda_M\lambda_R) - 4\theta^2(7+\lambda_M^4(1-\lambda_R)\lambda_R^2 + \lambda_M^3(2-8\lambda_R^3+4\lambda_R^2+6\lambda_R) - \lambda_M^2(9+5\lambda_R^3-21\lambda_R^2+19\lambda_R) + 2\lambda_M(6+\lambda_R^3+7\lambda_R^2-8\lambda_6) - 4\lambda_R^2 - 7\lambda_R) + 8\theta^3(1+\lambda_M)(3+\lambda_M^3(1-\lambda_R)\lambda_R^2 + \lambda_M^2(5-\lambda_R)\lambda_R^2 - 7\lambda_M\lambda_R - \lambda_M + \lambda_R) - \theta^4(1-\lambda_M)(1+\lambda_M)(1+\lambda_R)(3+\lambda_M^2(1+4\lambda_R-5\lambda_R^2) - 4\lambda_M(1+\lambda_R^2) - \lambda_R^2 + 4\lambda_R)$, *and* $B_3 \equiv 4(2-\lambda_M-\lambda_M\lambda_R) - 2\theta(1+\lambda_M)(1-\lambda_M\lambda_R) - \theta^2(1-\lambda_M)(1+\lambda_M)(1+\lambda_R)$, *Additionally, when* $0 < \lambda_M < 1$ *and* $0 < \lambda_R < \bar{\lambda}_R$ *are satisfied, all the outcomes are positive.*

Additionally, in contrast to the non-encroachment case, the strategic effect in retail market competition plays an important role in considering the effect of common ownership. Therefore, using Result 2, we consider the impact of ownership share on the quantity, wholesale price, profits, and surplus. While, differentiating outcomes by λ_R and λ_M, we examine the impact of common ownership. Because it is difficult to specify the understandable outcomes in $\partial\hat{\pi}_M^E/\partial\lambda_M$ and $\partial\hat{CS}/\partial\lambda_M$, we initially focus on other outcomes. Consequently, we obtain the following proposition in the asymmetric ownership share under the encroachment case:

Proposition 10. *The degree of ownership share has the following impact on the outcomes in the encroachment case.*

$$\frac{\partial\hat{q}_R^E}{\partial\lambda_M} > 0, \quad \frac{\partial\hat{q}_M^E}{\partial\lambda_M} < 0, \quad \frac{\partial\hat{w}^E}{\partial\lambda_R} > 0, \quad \frac{\partial\hat{w}^E}{\partial\lambda_M} < 0, \quad \frac{\partial\hat{\pi}_R^E}{\partial\lambda_R} < 0, \quad \frac{\partial\hat{\pi}_R^E}{\partial\lambda_M} > 0, \quad \frac{\partial\hat{\pi}_M^E}{\partial\lambda_R} > 0, \quad \frac{\partial\hat{TS}^E}{\partial\lambda_M} > 0.$$

Additionally, we obtain

$$\frac{\partial\hat{q}_R^E}{\partial\lambda_R} \gtreqless 0, \quad \lambda_M \gtreqless \alpha,$$
$$\frac{\partial\hat{q}_M^E}{\partial\lambda_R} \gtreqless 0, \quad \lambda_M \gtreqless \alpha,$$
$$\frac{\partial\hat{TS}^E}{\partial\lambda_R} \gtreqless 0, \quad \lambda_M \lesseqgtr \alpha,$$
$$\frac{\partial\hat{CS}^E}{\partial\lambda_R} > 0, \quad \textit{under } \lambda_M > \alpha \textit{ and } \theta > \beta,$$
$$\frac{\partial\hat{CS}^E}{\partial\lambda_R} < 0, \quad \textit{under } \lambda_M < \alpha \textit{ and } \theta < \beta,$$

where α *and* β *are defined in Appendix.*

Proof. See Appendix.

From this proposition, we can confirm that the effects of λ_R on w, π_R, and π_M do not differ between the non-encroachment and encroachment cases. Moreover, the effects of λ_M on q_R, w, π_R, and TS also do not differ between the non-encroachment and encroachment cases. However, depending on the parameters, λ_R and λ_M affect q_R, q_M, π_M, CS, and TS. In particular, in the non-encroachment case, the manufacturer does not set q_M because it does not open the direct channel. While $\hat{q}_M^E$ decreases as λ_M because the manufacturer does not obtain market share by emphasizing the retailer's profit. Depending on the economic environment, λ_R has a different effect.

In addition, we explore $\partial\hat{\pi}_M^E/\partial\lambda_M$ and $\partial\hat{CS}/\partial\lambda_M$ to improve our understanding. First, we confirm $\partial\hat{\pi}_M^E/\partial\lambda_M$ and obtain the following observations:

Observation 1. *The manufacturer's profit increases as* λ_M *increases under a specific economic environment.*

Proof. See Appendix.

Generally, an increase of λ_M means that the manufacturer strongly emphasizes the retailer's profit; therefore, the manufacturer enhances the retail market by reducing the quantity or the wholesale price to enhance the retailer's profit. However, in a specific case, by reducing the quantity, both firms can engage in tacit collusion in the retail market. On the other hand, because the manufacturer reduces the wholesale price to improve the retailer's profit and the retailer can enhance the quantity, the total surplus improves as λ_M increases.

Next, we consider $\partial\hat{CS}^E/\partial\lambda_M$ and obtain the following observations:

Observation 2. *The consumer surplus increases as* λ_M *increases in a specific economic environment.*

Proof. See Appendix.

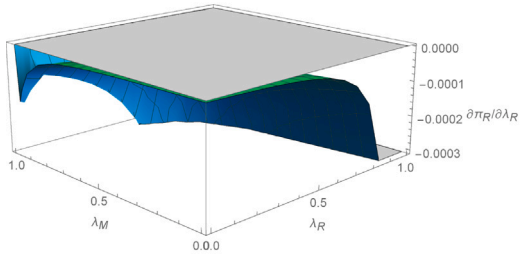

Fig. 4. The effect of λ_R and λ_M on $\partial\pi_R/\partial\lambda_R$ under $a = 1$ and $\theta = 99/100$.

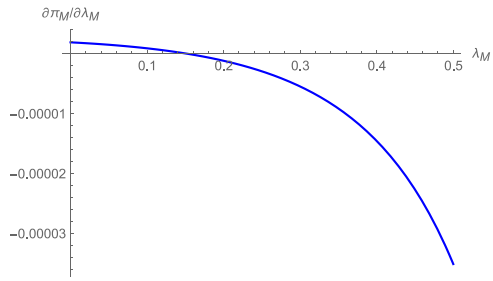

Fig. 5. The effect of λ_M on $\partial\pi_M/\partial\lambda_M$ under $a = 1$ and $\theta = 99/100$.

This observation indicates that, while the manufacturer enhances common ownership, it may yield a positive outcome for consumer surplus. Additionally, while consumer surplus decreases as λ_M increases in a specific case, tacit collusion improves the total profits of the supply chain parties. This opposite effect leads to an improvement in total surplus.

Finally, we analyze whether manufacturer encroachment under common ownership improves the retailer's profit. Based on this analysis, we obtain the following proposition.

Proposition 11. *In a specific economic environment, manufacturer encroachment improves the retailer's profit. Formally,* $\Delta\hat{V}_R \equiv \hat{V}_R^E - \hat{V}_R^{NE} > 0$ *for a specific case.*

Proof. See Appendix.

From this additional analysis, we can confirm that the main results in the prior sections are obtained. Therefore, under asymmetric ownership share, while we obtain the complex result, we can confirm that our main result is robust. However, an asymmetric case helps to understand the effect of common ownership among supply chain parties with manufacturer encroachment. This robustness check encourages us to suggest our implications for decision-making and regulation practice.

Lastly, we consider the optimal choice of λ_R and λ_M to maximize the profits π_R and π_M respectively. Under the non-encroachment case, because we can confirm that both firms' profit decreases as their own share increases from Proposition 9, both firms do not enhance the level of λ_i to improve the profit. Consequently, under the non-encroachment case, both firms choose $\lambda_i = 0$ to maximize their profits.

On the other hand, under the encroachment case, we cannot obtain the explicit outcome for the complexity. Therefore, we specify θ and propose the important outcomes numerically in this analysis. For example, under $\theta = 99/100$, we observe that $\lambda_M \simeq 2.40864$ and $\lambda_R \simeq 0.289482$ are obtained as outcomes of maximization problems, which satisfy the second-order condition. However, the above outcomes do not satisfy the conditions of $0 < \lambda_M < 1$ and $0 < \lambda_R < 1$.

To consider this problem, we confirm whether the corner solution is optimal for both firms using the figure. First, we consider $\partial\pi_R/\partial\lambda_R$ because the owner of the retailer decides the level of ownership to maximize the profit. Considering $\partial\pi_R/\partial\lambda_R$, we draw the figure with respect to λ_R and λ_M in Fig. 4. In this figure, the gray surface denotes the region $\partial\pi_R/\partial\lambda_R = 0$, and we can confirm that $\partial\pi_R/\partial\lambda_R = 0$ is always negative in under $a = 1$ and $\theta = 99/100$. In other words, π_R decreases as λ_R increases, and therefore, the owner of the retailer does not enhance λ_R to maximize the profit. This implies that the owner chooses $\lambda_R = 0$ in this case.

On the other hand, given that the owner chooses $\lambda_R = 0$, the manufacturer considers the optimal λ_M in this case. To consider the effect of λ_M on $\partial\pi_M/\partial\lambda_M$, we present Fig. 5. In this figure, we can find that, depending on the level of λ_M, $\partial\pi_M/\partial\lambda_M > 0$ and $\partial\pi_M/\partial\lambda_M < 0$ are obtained. In other words, there is an optimal level of λ_M as an interior solution under $a = 1$ and $\theta = 99/100$ to maximize π_M. We solve $\partial\pi_M/\partial\lambda_M = 0$ to consider the maximization problem, we obtain $\lambda_M \simeq 0.153$ as an optimal λ_M. Because, under $\lambda_M > 0.153$, π_M decreases as λ_M increases, and, under $\lambda_M < 0.153$, π_M increases as λ_M increases, $\lambda \simeq 0.153$ maximizes the profit of the manufacturer.

Table 2
Outcomes under non-encroachment and encroachment case with a direct selling cost.

	Non-encroachment case (NE)	Encroachment case (E)
$\tilde{w}$	$\frac{1}{(1-\lambda)(2+\lambda)}$	$\frac{5+3\lambda-c_M(1+\lambda)}{2(5+3\lambda)}$
$\tilde{q}_R$	$\frac{1+\lambda}{2(2+\lambda)}$	$\frac{2c_M(1+\lambda)}{(1-\lambda)(5+3\lambda)}$
$\tilde{q}_M$	–	$\frac{(1-\lambda)(5+3\lambda)-c_M(7+2\lambda-\lambda^2)}{2(1-\lambda)(5+3\lambda)}$
$\tilde{p}_R$	$\frac{3+\lambda}{2(2+\lambda)}$	$\frac{5+3\lambda+c_M(3+\lambda)}{3(5+3\lambda)}$
$\tilde{p}_M$	–	$\frac{5+3\lambda+c_M(3+\lambda)}{3(5+3\lambda)}$
$\tilde{\pi}_M$	$\frac{1+\lambda}{2(1-\lambda)(2+\lambda)^2}$	$\frac{1}{4}\left[\frac{c_M^2(45+41\lambda-\lambda^2-5\lambda^3)}{(1-\lambda)(5+3\lambda)^2}+1-2c_M\right]$
$\tilde{\pi}_R$	$\frac{(1+\lambda)(1-2\lambda-\lambda^2)}{4(1-\lambda)(2+\lambda)^2}$	$\frac{2c_M^2(1+\lambda)(2+\lambda)}{(1-\lambda)(5+3\lambda)^2}$
$\tilde{V}_M$	$\frac{(1+\lambda)^2}{4(2+\lambda)}$	$\frac{(1-\lambda)(5+3\lambda)-2c_M(1-\lambda)(5+3\lambda)+c_M^2(3+\lambda)^2}{4(1-\lambda)(5+3\lambda)}$
$\tilde{V}_R$	$\frac{(1+\lambda)^2}{4(2+\lambda)^2}$	$\frac{1}{4}\left[\frac{c_M^2(16+69\lambda+49\lambda^2-\lambda^3-5\lambda^4)}{(1-\lambda)(5+3\lambda)^2}+\lambda(1-2c_M)\right]$
$\tilde{CS}$	$\frac{(1+\lambda)^2}{8(2+\lambda)^2}$	$\frac{[5+3\lambda-c_M(3+\lambda)]^2}{8(5+3\lambda)^2}$
$\tilde{TS}$	$\frac{(1+\lambda)^2}{8(2+\lambda)^2}$	$\frac{3(1-\lambda)(5+3\lambda)^2-2c_M(1-\lambda)(5+3\lambda)(13+7\lambda)+c_M^2(131+127\lambda+9\lambda^2-11\lambda^3)}{8(1-\lambda)(5+3\lambda)^2}$

Assuming the endogenous ownership, we find that the common ownership may arise in equilibrium. This result also indicates that it becomes a good consequence for the supply chain to have a small share of the retailer. One can infer that while this result may suggest vertical integration to the manufacturer, the existence of the decision rights for the retailer plays an important role in enhancing the supply chain-wide profit.

However, under the common ownership case, one can consider that owners hope to enhance the supply chain-wide profit because firms partially have common owners in our assumption. In the future study, one can analyze the case to maximize the supply chain-wide profit with common ownership.

5.4. Additional cost on a direct channel

Finally, following Arya et al. (2007), we analyze the case in which the manufacturer is confronted with selling costs in a direct channel. Given these circumstances, the manufacturer's profit is defined as $\tilde{\pi}_M = (p_M - c_M)q_M + wq_R$, where $\sim$ denotes the case in which the manufacturer faces selling costs in the direct channel and c_M (> 0) represents additional selling costs in the direct channel. In addition, we normalize $a = 1\,(> c_M)$ and $\theta = 1$ for simplicity. While the degree of product differentiation has an important role to consider the main model, because we can demonstrate the robustness of our main results and our purpose of this section is to show the robustness of our results, we normalize θ as 1. The other settings were the same as those in the basic model. Solving this case using backward induction, we obtain the outcomes proposed in Table 2. When $\lambda < \sqrt{2}-1$ and $c_M < (1-\lambda)(5+3\lambda)/(7+2\lambda-\lambda^2)$ are satisfied, all outcomes are positive. Therefore, we assume $\lambda < \sqrt{2}-1$ and $c_M < (1-\lambda)(5+3\lambda)/(7+2\lambda-\lambda^2)$.

These outcomes demonstrate that direct selling costs play an important role in firms' strategies and profits. In particular, wholesale prices decrease as c_M increases, and market prices increase as c_M increases. In addition, the retailer's quantity, $\tilde{q}_R^E$, increases as c_M increases, and the manufacturer's quantity, $\tilde{q}_M^E$, decreases as c_M increases. This result indicates that direct selling costs improve the retailer's market share and that the manufacturer is adversely affected by direct selling costs.

First, we examine the profits and encroachment decisions of manufacturers. We define $\Delta\tilde{V}_M \equiv \tilde{V}_M^E - \tilde{V}_M^{NE}$ and obtain the following outcome.

$$\Delta\tilde{V}_M = \frac{(1-\lambda)(5+3\lambda)\left(1-\lambda-\lambda^2\right) - 2c_M(1-\lambda)(2+\lambda)(5+3\lambda) + c_M^2(2+\lambda)(3+\lambda)^2}{4(1-\lambda)(2+\lambda)(5+3\lambda)}. \tag{37}$$

When $(1-\lambda)(5+3\lambda)/(3+\lambda)^2 < c_M < \sqrt{(1+\lambda)^4(1-\lambda)(5+3\lambda)/(2+\lambda)(3+\lambda)^4}$, $\Delta\tilde{V}_M < 0$ holds. Therefore, in this case, $\tilde{V}_M^E < \tilde{V}_M^{NE}$ is obtained. The manufacturer does not open direct channels in the retail market. We summarize this result in the following proposition.

Proposition 12. *In a specific economic environment, the manufacturer does not encroach on the retail market with direct selling costs. Formally, $\Delta\tilde{V}_M < 0$ holds when $(1-\lambda)(5+3\lambda)/(3+\lambda)^2 < c_M < \sqrt{(1+\lambda)^4(1-\lambda)(5+3\lambda)/(2+\lambda)(3+\lambda)^4}$.*

The negative effect of direct selling costs on manufacturer quantity leads to this result. When c_M is sufficiently large, the manufacturer reduces the quantity supplied to the product market through the direct selling costs. Therefore, encroachment is not profitable for the manufacturers. Fig. 6 shows the regions where $\Delta\tilde{V}_M > 0$ and $\Delta\tilde{V}_M < 0$. This figure implies that encroachment is not profitable for the manufacturer under conditions of high c_M and λ.

In addition, we consider the effects of manufacturer encroachment on retailer profit. After defining $\Delta\tilde{V}_R \equiv \tilde{V}_R^E - \tilde{V}_R^{NE}$, we obtain the following outcome.

$$\Delta\tilde{V}_R = \frac{-(5+3\lambda)^2(1-\lambda)\left(1-2\lambda-3\lambda^2-\lambda^3\right) - 2c_M\lambda(1-\lambda)(2+\lambda)^2(5+3\lambda)^2 + c_M^2 B_2(2+\lambda)^2}{4(1-\lambda)(\lambda+2)^2(3\lambda+5)^2}, \tag{38}$$

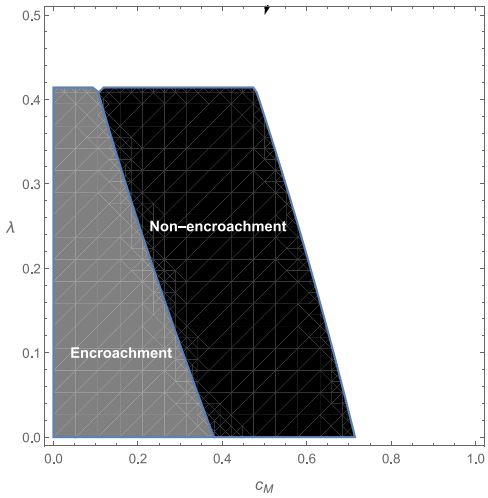

Fig. 6. Regions of $\Delta\tilde{V}_M > 0$ and $\Delta\tilde{V}_M < 0$.

where $B_2 \equiv 16+69\lambda+49\lambda^2-\lambda^3-5\lambda^4$, We can demonstrate that a case exists in which $\Delta\tilde{V}_R > 0$ under a specific c_M. In addition, this result can be specified more easily than when no direct selling costs are incurred. This result implies the bright side of manufacturer encroachment on the retailer's profit with direct selling costs, similar to Arya et al. (2007). We summarize this result in the following proposition:

Proposition 13. *Under specific conditions, manufacturer encroachment enhances the retailer's profit through direct selling costs. Formally, $\Delta\tilde{V}_R > 0$ for a specific c_M.*

Proof. See Appendix.

Next, we consider welfare effects. Under these conditions, $\Delta\tilde{CS} \equiv \tilde{CS}^E - \tilde{CS}^{NE}$ is computed as

$$\Delta\tilde{CS} = \frac{\left[5+3\lambda-c_M(2+\lambda)(3+\lambda)\right]\left[(3+2\lambda)(5+3\lambda)-c_M(2+\lambda)(3+\lambda)\right]}{8(2+\lambda)^2(5+3\lambda)^2} > 0. \tag{39}$$

Therefore, in this case, ΔCS is always positive when $\lambda < \sqrt{2}-1$ and $c_M < (1-\lambda)(5+3\lambda)/(7+2\lambda-\lambda^2)$.

In addition, $\Delta\tilde{TS} \equiv \tilde{TS}^E - \tilde{TS}^{NE}$ is computed as:

$$\Delta\tilde{TS} = \frac{(1-\lambda)(5+2\lambda)(5+3\lambda)^2 - 2c_M(1-\lambda)(5+3\lambda)(13+7\lambda)(2+\lambda)^2 + c_M^2 B_3(2+\lambda)^2}{8(1-\lambda)(2+\lambda)^2(5+3\lambda)^2}, \tag{40}$$

where $B_3 \equiv 131+127\lambda+9\lambda^2-11\lambda^3$, From this outcome, $\Delta\tilde{TS} < 0$ is obtained under specific economic conditions. We summarize this result in the following proposition:

Proposition 14. *Under specific conditions, manufacturer encroachment adversely affects the total surplus (i.e., $\Delta\tilde{TS} < 0$).*

Proof. See Appendix.

This proposition implies that whereas manufacturer encroachment improves consumer surplus, total surplus is adversely affected by manufacturer encroachment because the effects of opening a direct channel with direct selling costs on profits are strongly negative. Therefore, although it is profitable for a consumer to open a direct channel, manufacturer encroachment is not profitable. Consequently, manufacturer encroachment adversely affects the total surplus. However, this is not a unique result when considering common ownership. Although common ownership affects the threshold of $\Delta\tilde{TS} < 0$, it plays no important role in the consideration of this result.

6. Conclusion

This study examines the effects of common ownership on supply chain management with the manufacturer's encroachment decision. The results of our analyses demonstrate that given a large share of ownership by the manufacturer, the retailer's profit is improved by manufacturer encroachment. Additionally, we demonstrate that increasing ownership share improves the total surplus in a specific case. In our additional analysis, we confirm the robustness and extensibility of our model. In particular, an asymmetric ownership share and the manufacturer's direct selling cost have an important role in practice.

These results constitute important contributions to supply chain management practice and financial economics studies. First, the result obtained for the retailer's profit suggests that common ownership directly improves the retailer's profit and indirectly enhances the profit in the encroachment case. This result implies that the retailer should aggressively engage in common ownership between the retailer and manufacturer. Second, we can ascertain some positive aspects of common ownership regarding the total surplus. Therefore, for supply chain management, society should accept common ownership by manufacturers and retailers.

We can consider the following future research based on our model. First, for our main model, we assume a monopoly case by using one supply chain. Therefore, future research should include analyses incorporating the assumptions of a competitor in a product market. Second, while this study considers only quantity competition, future research can consider cases that include price competition. Third, recently, to explore optimal decision-making by supply chain parties, several studies assume that other kinds of agents have a specific utility. In particular, risk-averse agents are supposed to analyze the supply chain coordination (e.g., Matsui, 2019). Therefore, in future research, one can consider the effect of risk-averse agents under common ownership among supply chain parties. While one can consider the above future studies based on our model, there are beyond our scope, and we propose a basic analysis of the impact of common ownership on the dual-channel supply chain.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and material

Not applicable.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.iref.2024.103596.

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